

Paper OOL-9: Protocell Integration, Replication, and Error Thresholds: The First Heritable Container Gate

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Abstract

This paper is the integration gate. A protocell claim requires bounded copying error, retained internal chemistry, coupled energy throughput, and a boundary whose persistence covaries with its internal state. The paper combines Eigen–Orgel error-threshold logic with a conservative protocell diagnostic and explicitly reports numerical fragility where it appears. That restraint is intentional: Paper 9 defines what must be true before Paper 10 can introduce parasitic exploitation.

1 Introduction

Papers 1–8 build the prerequisite ladder: thermodynamic framing, geochemical energy and transport, oscillating polymerization, autocatalytic closure, and adaptive boundaries. A protocell requires these pieces to become coupled. Replication without a boundary may preserve molecular information but does not necessarily create a lineage. A boundary without inherited internal chemistry can divide physically without biological heredity.

Paper 9 therefore asks two linked questions:

1. When does copying preserve information rather than dissolve into error?
2. When does copying affect the persistence and division of the container that carries it?

2 Template Growth and Replication Gain

For a schematic template T and monomer pool M_i ,

$$T + M_i \rightarrow TM_i, \quad R_{\text{rep}} = k_{\text{rep}} c_T c_M. \quad (1)$$

Transport-limited monomer supply can be represented phenomenologically as

$$\mathbf{J}_M = -\frac{1}{C} \nabla c_M. \quad (2)$$

A well-mixed caricature gives

$$\frac{dc_T}{dt} = k_{\text{rep}}c_Tc_M - \frac{1}{CL^2}c_T = c_T \left(k_{\text{rep}}c_M - \frac{1}{CL^2} \right). \quad (3)$$

Define the replication gain

$$\Gamma_{\text{rep}} = k_{\text{rep}}c_MCL^2. \quad (4)$$

The condition $\Gamma_{\text{rep}} > 1$ is only a linearized growth condition. It is not sufficient for life.

3 Error Threshold

If per-step fidelity is q across n additions, the simplest master-sequence survival approximation is

$$Q = q^n, \quad \epsilon = 1 - Q. \quad (5)$$

Information persists only if replication gain outruns error catastrophe. A schematic compatibility condition is

$$\Gamma_{\text{rep}}Q > 1, \quad (6)$$

with the usual warning that real prebiotic copying is not a clean independent-site process [4, 2].

The CDFD hypothesis is that fidelity, monomer activation, retention, and damage are coupled to energy throughput. A useful bookkeeping score is

$$\mathcal{H}_{\text{copy}} = \Gamma_{\text{rep}}QR_{\text{mono}}R_{\text{ret}}R_{\text{damage}}^{-1}, \quad (7)$$

where R_{mono} is activated-monomer availability, R_{ret} is retention of templates and products, and R_{damage} collects hydrolysis, photodamage, side reactions, and strand loss. This is a checklist, not a calibrated law.

4 Quasispecies Logic Versus Protocell Logic

The quasispecies threshold and the protocell threshold are related but not identical. The first asks whether a sequence distribution preserves information. The second asks whether a bounded domain inherits a coupled chemistry that affects its growth or persistence.

The protocell condition can be stated as a covariance requirement:

$$\text{Cov}(\text{copied internal state}, \text{container persistence}) > 0. \quad (8)$$

Without this covariance, selection may act on free templates or on empty containers, but not yet on a protocell lineage.

5 Integrated Protocell Scaffold

An integrated toy protocell has state variables such as

$$U = \{c_i, \Phi_e, \Phi_H, C_i, \dots\}, \quad (9)$$

evolving through transport, reaction, and constraint laws [6]. A minimal transport skeleton is

$$\frac{\partial c_i}{\partial t} = \nabla \cdot \left(\frac{1}{C_i} \nabla c_i \right) + R_i(c, \Phi_e, \Phi_H), \quad (10)$$

with potentials satisfying analogous updates:

$$\frac{\partial \Phi_e}{\partial t} = \nabla \cdot \left(\frac{1}{C_e} \nabla \Phi_e \right) + S_e, \quad (11)$$

$$\frac{\partial \Phi_H}{\partial t} = \nabla \cdot \left(\frac{1}{C_H} \nabla \Phi_H \right) + S_H. \quad (12)$$

Flux-coupled reactions can be written as

$$\mathbf{J}_e = -\frac{1}{C_e} \nabla \Phi_e, \quad \mathbf{J}_H = -\frac{1}{C_H} \nabla \Phi_H, \quad (13)$$

$$R_i = f_i(c) g(|\mathbf{J}_e \cdot \mathbf{J}_H|), \quad (14)$$

for nonnegative g .

6 Numerical Audit

The diagnostic for this paper combines a reduced error-threshold sweep with a bounded two-dimensional protocell update. It should be read as a numerical stress test, not as validation. The reduced diagnostic reported:

- $\Phi = 0.2$: final master-sequence fitness 0.367,
- $\Phi = 0.5$: final master-sequence fitness 0.083,
- $\Phi = 1.0$: final master-sequence fitness 0.014,
- $\Phi = 5.0$: final master-sequence fitness 1.000,
- $\Phi = 10.0$: final master-sequence fitness 1.000.

The non-monotonic low-to-intermediate response rules out a single universal critical flux in this reduced diagnostic. The correct claim is windowed: copying benefits from energy throughput only when activation, retention, and damage remain bounded.

The final release replaces the earlier unbounded explicit autocatalytic update with saturated kinetics and bounded adaptive S, M_s updates. Under those release parameters, the two-dimensional protocell diagnostic remains finite for the full run. The result should still be read conservatively: the simulated boundary protects a low internal Ψ_s region while the exterior

experiences large drive, which is a stress-test picture of compartmental buffering rather than a demonstration of real protocell reproduction.

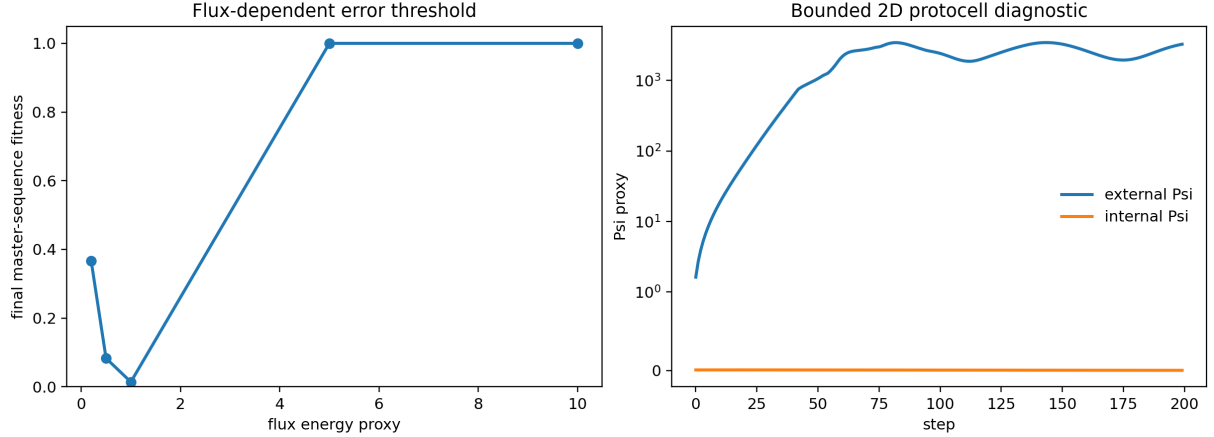


Figure 1: Merged error-threshold/protocell diagnostic. The left panel tests flux-dependent master-sequence persistence. The right panel shows a bounded two-dimensional protocell stress test with finite internal and external adaptive-ratio traces. The figure is a mechanism diagnostic, not a successful protocell simulation.

7 Hardening Roadmap

The release implementation now uses saturated kinetics such as

$$R_{\text{auto}} = \frac{k\Phi^2}{K + \Phi^2} - \ell\Phi, \quad (15)$$

and verifies boundedness in the generated summary JSON. The next hardening step is stronger: timestep refinement, grid-size refinement, and comparison against operator-split or implicit updates for coupled transport and reaction. The model should only be promoted from stress test to prediction if those refinement checks preserve the qualitative result.

8 Experimental Targets

The model protocell literature gives concrete targets: template-directed genetic-polymer synthesis in fatty-acid vesicles [3], nonenzymatic RNA synthesis inside model protocells [1], multi-step nonenzymatic primer-extension chemistry [7], and recent water-compatible thioester-mediated RNA aminoacylation/peptidyl-RNA chemistry [5]. CDFD should therefore be tested by a coupled score,

$$\mathcal{R}_{\text{proto}} = \Gamma_{\text{rep}} Q R_{\text{retention}} R_{\text{exchange}} R_{\text{division}}, \quad (16)$$

not by replication alone.

9 Conclusion

Paper 9 is the integration gate. It establishes that bounded copying, retained internal chemistry, coupled energy flow, and heritable containers must be tested together. The current numerical status is deliberately conservative: the reduced copying toy is informative, while the two-dimensional protocell implementation is a bounded stress test that still needs refinement before it can be treated as a prediction. Paper 10 can only discuss parasitic exploitation after this replication/container machinery exists.

10 Reproducibility Note

The companion script `supplementary_ool_09.py` writes the figure `outputs/paper_09/error_threshold_and_protocell.png` plus CSV/JSON summaries. No notebook is required.

References

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